

TROUBLESHOOTING ASS.

M	T	W	T	F	S	S
Page No.:						YOUVA
Date:						

Day 1(1): List 3 real-life scenarios where a multi-meter would be used to troubleshoot a computer or smartphone issue, & explain in one line how the multi-meter helps in each case.

→ Power adapter failure: Check if the charger outputs correct voltage.

Broken charging port: Test line continuity for short circuit or damaged pins.

Dead Battery: Measure voltage to determine if the battery holds a charge.

Day 1(2): Watch a short YouTube video demo of a POST card in action, then write a brief summary.

→ The POST diagnostic card displays two digit hexadecimal codes across two seven-segment LED screens, along-side voltage & system signal LED.

Day 1(3): Download & install any free software diagnostic tool, run it on laptop or desktop.

Day 1(4): compare the use of a multimeter, POST card, & software diagnostics by creating a table with 3 columns

42(1):

Tool	What it checks	Ex. Problem detected
Multimeter	Voltage, current, resistance	Dead charger or not circuit
POST card	low-level BIOS boot codes.	Failed RAM or CPU initialization.
Software Diagnostic	Component S.M.A.R.T health & temperatures	Drive failure or overheating

42

→

42

DAY 2

Day 2(1): List 5 common hardware issues you might encounter in a desktop PC & write the physical symptoms.

→ **Faulty RAM**: No display with continuous beep codes or random BSODs.

Failing PSU: PC won't power on or turns off instantly under load.

CPU overheating: System shuts down after a few minutes with loud fans.

Failing storage: "No boot device found" error or freezing at boot.

Dead GPU: screen artifacts or "No signal" on monitor.

Day 2(2): Given this scenario: A laptop is not charging even when plugged in.

- (1) **Outlet & cable**: check socket power & cord damage
- (2) **Laptop port**: check for dirt or broken pins.
- (3) **Test adapter**: measure charger voltage with a multimeter.
- (4) **Static Reset**: Remove battery, hold power for 30s.
- (5) **Check battery**: Inspect for swelling or bad contact.

Day 2(3): Use a multimeter to test the voltage output of a standard USB phone charger.

→ Measured voltage: 5.12 V DC
Expected Range: 4.75 V to 5.25 V DC
Status: PASS - within expected range

Day 2(4): A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail.

→ Jump PSU: connect 24pin Green & Black pin with a paperclip to turn it on.

Set multimeter: dial to DC voltage

Probe: Put black probe in black pin & red probe in red pin.

Measured voltage: 12.15 V DC.